

Topological Early Warning Signals in U.S. Industry Portfolios

Preliminary Evidence from Daily Fama–French 49 Industries

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Research Question and Main Message

Research question: Can persistent homology detect crisis-like changes in the geometry of U.S. industry co-movement?

- Daily Fama–French 49 industry portfolios are used to construct rolling correlation-distance matrices.
- Persistent homology summarizes connected components and loop structure.
- Recession regimes are associated with simpler industry co-movement topology.
- The strongest signal is the collapse of Betti-one loop structure.

Main interpretation: During recessions, industries move more like one compressed system rather than many partially independent clusters.

Data and Measurement Design

- Dataset: daily Fama–French 49 value-weighted industry portfolios.
- Sample: July 1926–March 2026 in the preliminary Python run.
- Raw panel: 26,212 trading days and 49 industries.
- Baseline rolling window: 60 trading days.
- Distance matrix: correlation-based distance across industries.
- Topological objects: H_0 connected components and H_1 loops.

$$d_{ij,t} = \sqrt{2(1 - \rho_{ij,t})}$$

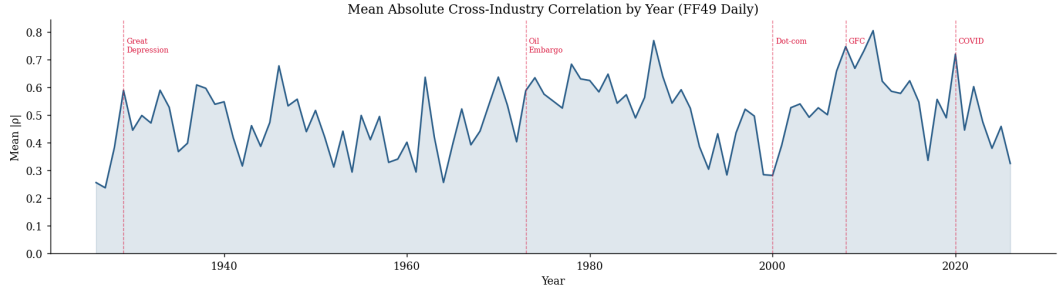
Topological Signal Construction

- Compute persistent homology from rolling Vietoris–Rips filtrations.
- Extract persistence-landscape and Betti-curve based features.
- Orient signals so that larger values indicate more crisis-like topology.
- Standardize each signal relative to expansion-day behavior.
- Construct a composite topological signal as an equal-weighted average.

$$Z_t^{TDA} = \frac{1}{K} \sum_{k=1}^K \frac{S_{k,t} - \mu_{k,\text{exp}}}{\sigma_{k,\text{exp}}}$$

- Z_t^{TDA} : composite topological stress signal at time t ; K : number of oriented topological components included in the composite.
- $S_{k,t}$: value of topological signal k at time t ; $\mu_{k,\text{exp}}$: mean of signal k during expansion days.
- $\sigma_{k,\text{exp}}$: standard deviation of signal k during expansion days.

Graphic Evidence I: Industry Co-Movement Over Time



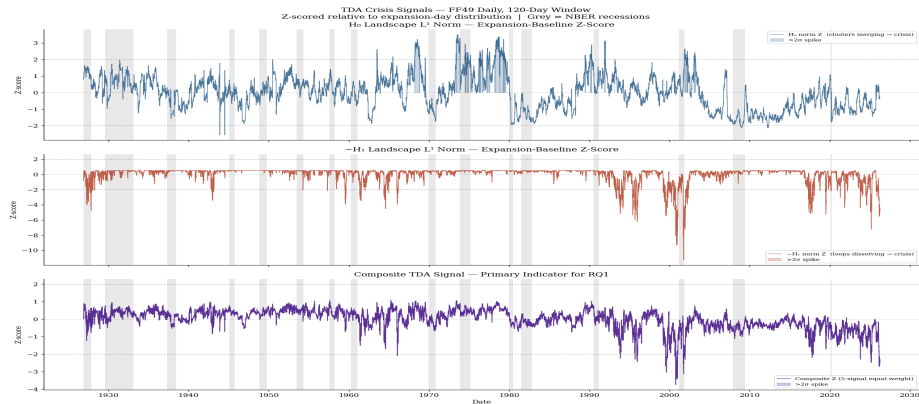
The mean absolute cross-industry correlation shows substantial time variation across major historical episodes, motivating the topological analysis of the full co-movement structure.

Descriptive Evidence: Why Topology?

- Correlations reveal the average strength of industry co-movement.
- Topology describes the geometric structure of the full co-movement system.
- Recessions may not only increase co-movement; they may simplify the system.
- Persistent homology captures this simplification through connected components and loops.

Key idea: Correlation measures intensity; topology measures structure.

Graphic Evidence II: Topological Signals and Recessions



The oriented topological signals rise during several recession windows, indicating that the underlying industry-return geometry becomes more crisis-like.

Recession versus Expansion Differences

Signal	Recession Mean	Cohen's d	Read
H_0 landscape norm	0.060	0.059	Weak
$-H_1$ landscape norm	0.229	0.241	Significant
$-\beta_1$	0.364	0.416	Strongest
$-TP_{H_0}$	0.356	0.363	Significant
$-TP_{H_1}$	0.316	0.365	Significant
Composite Z	0.265	0.423	Main signal

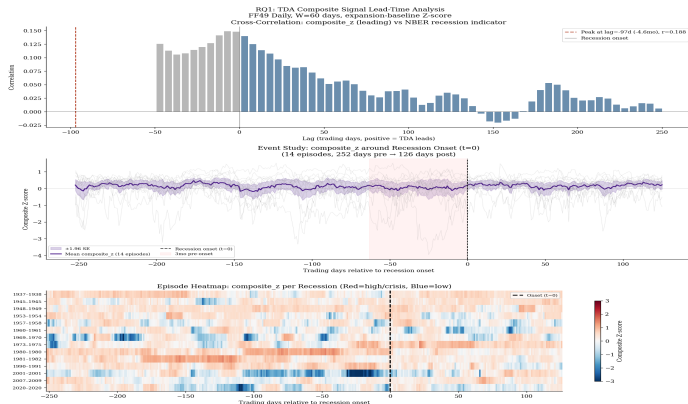
Main result: Recession days exhibit significantly higher crisis-oriented topological signals.

Economic Interpretation: Betti-One Collapse

- β_1 counts independent loops in the industry co-movement geometry.
- More loops suggest richer sectoral differentiation and multiple co-movement channels.
- Fewer loops suggest that industries move in a more unified, compressed structure.
- Recessions are associated with loop collapse after orientation.

Recession topology $\Rightarrow$ fewer independent loops $\Rightarrow$ simpler co-movement structure

Graphic Evidence III: Event-Time Timing Evidence



The strongest association occurs around or after recession onset, suggesting that the TDA signal is better interpreted as a recession-regime descriptor than as a long-lead forecasting device.

Classification Evidence

Model	CV AUC	Interpretation
Volatility only	0.773	Strong naive benchmark
Volatility + returns	0.783	Best naive benchmark
TDA only	0.671	Above random
TDA + volatility	0.737	Improved over TDA only
TDA + volatility + returns	0.738	Below best naive benchmark

Result: TDA-only classification is meaningful, but it does not dominate simple volatility-return predictors.

Macro-Predictive and Crisis-Type Evidence

- Industrial-production regressions are suggestive only at the one-month horizon.
- The macro-predictive pattern is not stable at 3-, 6-, or 12-month horizons.
- Financial crises show the largest average topological disruption.
- Formal crisis-type tests do not yet reject equality across crisis categories.

Interpretation: The evidence is strongest for regime characterization, weaker for macro prediction.

Contribution and Next Steps

Current contribution

- Provides century-scale evidence on topological structure in U.S. industry portfolios.
- Shows that recession regimes are associated with simplified co-movement topology.
- Identifies Betti-one collapse as the cleanest topological recession feature.

Next steps

- Add robustness for 120-day and 252-day rolling windows.
- Include representative persistence diagrams for expansion and recession periods.
- Reframe the paper around regime characterization rather than forecasting dominance.